

## M4: Text Mining Basics

CAS Data Engineering and Applied Data Science 1

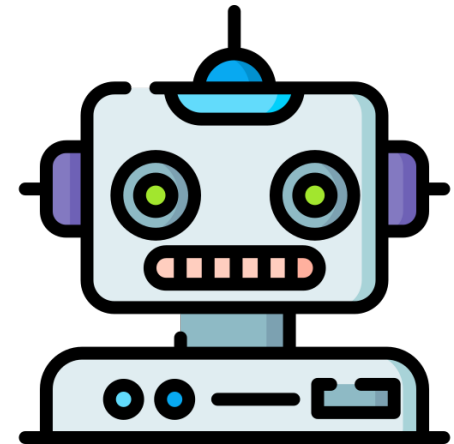
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## Upcoming Lectures

- 26.05: Supervised and Unsupervised Learning
- 02.06: Supervised and Unsupervised Learning II
- **03.06: Text Mining Basics**

# Agenda

- Introduction to Text Analysis
- Text Basics
- Text Similarity
- What is ChatGPT?
- Prompt Engineering
  - Group Work
- Extension + Presentations from "ML in the Wild"

## Recap :)

- Regression
  - Motivation
  - How to measure errors?
- Classification
  - Motivation
  - How to measure errors?
  - What to be careful about?
- Evaluation workflows
  - Motivation + Simple workflow
- Clustering
  - Motivation
  - K-Means + Agglomerative clustering
- Dimensionality reduction

# Introduction

# What is Text Analysis?

- Series of operations completed by one or more pieces of software on a sample of written human language (text), with the **goal of extracting useful information**
- These analyses can be performed by humans, but in a (much) longer period of time
- Part of the larger field of human language technology (HLT) or **natural language processing (NLP)**
  - Text Analysis: from language to other meaningful information
  - HLT / NLP also include text generation and interactive systems

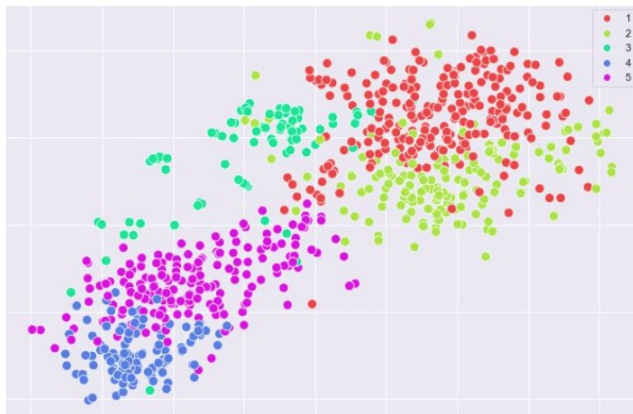


# Teaser



Are you looking for an opportunity to not just develop software, but to help build a system that will utilize **artificial intelligence SKILL** ( **AI SKILL** ) and **machine learning SKILL** to provide personalized human performance and training feedback for a US Air Force customer? As a **full-stack developer JOBTITLE**, you can solve a problem with a complete end-to-end solution in a fast-paced, **agile environment SKILL**. We're looking for a **developer JOBTITLE** with the skills needed to build an **AI SKILL** and **deep learning SKILL** (DL) solution that will interact with individuals to pers

safety, training, proficiency, and tactics for pilots and **intelligence SKILL** professionals, from vision to production-ready system.



# Applications of Text Analysis

## ANALYSIS

Spell checkers, incl. grammar  
Keyword extraction  
Authorship attribution  
Document retrieval  
Text classification  
Text mining  
Sentiment analysis  
Content-based recommendation

## TEXT ANALYSIS + GENERATION

Machine translation  
Automatic question answering  
Automatic summarization

## TEXT GENERATION

Generating reports from databases  
Generating weather forecasts

## ANALYSIS + GENERATION + INTERACTION

Dialogue systems  
Assistive technology: teaching, writing

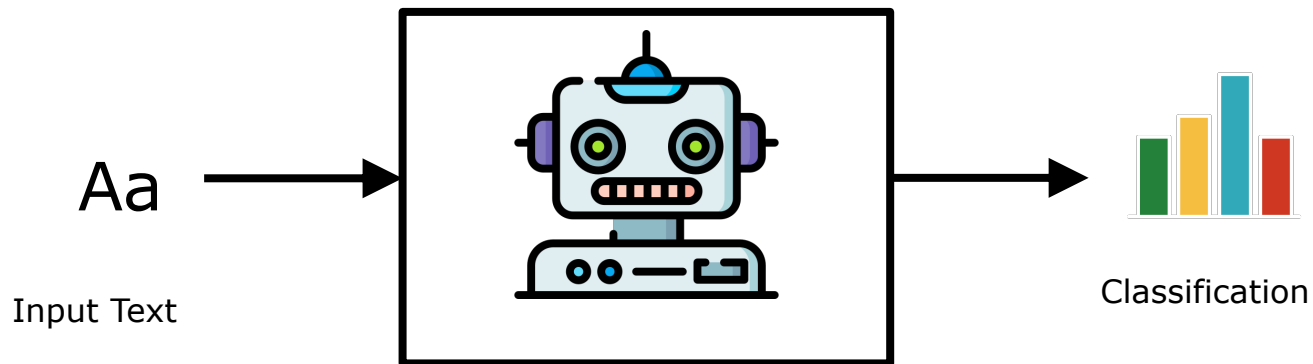


# Text Basics

# Basic Concepts for Text Analysis

Machine learning is a powerful tool because many text analysis problems are classification based

- Assign an item to one of the possible classes, given the features
- **Feature extraction + classification**
- Regression tasks are also very common



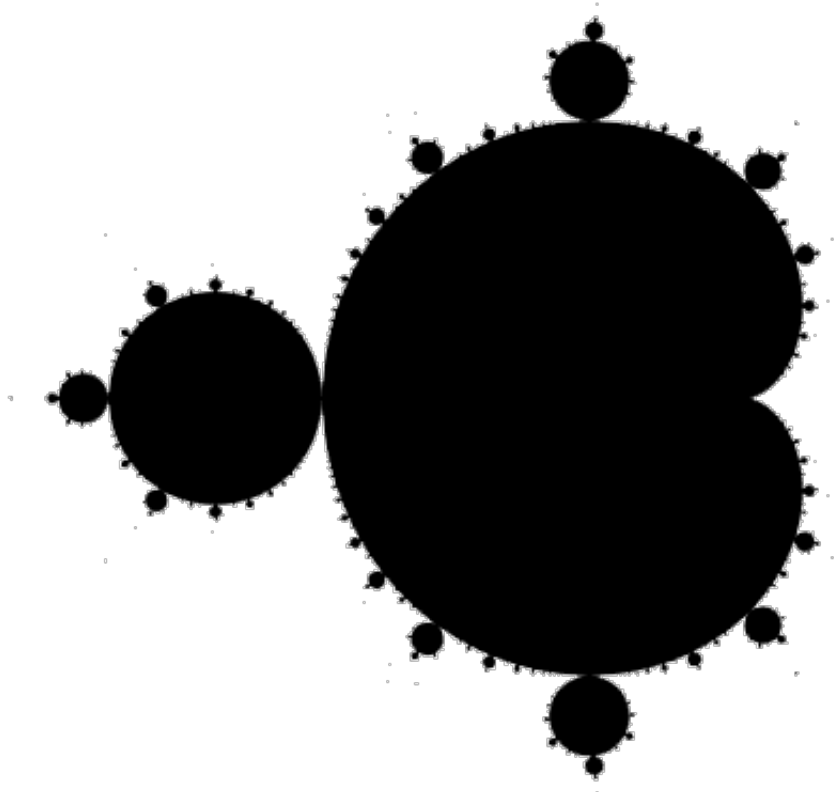
# Data for Text Analysis

Text analysis using machine learning requires **large amounts of training data** (and smaller amounts of test data)

- Finding suitable data is often a bottleneck (expensive, limited rights)

An **annotated corpus** typically contains:

- Selection of texts based on explicit criteria (e.g., genre, topic, time period, dialect)
- Metadata: author(s), date, source, title, sectioning, etc.
- Annotations in a more or less standardized format (e.g. XML, JSON, line-based (e.g. word + annotation), dictionary-style)



# TextBlob

A Python NLP library that sits on the mighty shoulders of NLTK and Pattern

Easy accessible and very well suited for fast prototyping ...

# Lexical Analysis

Goal: to understand word forms

- Tokenization: breaking a text into word tokens
- Lemmatization: finding the base form of each word token (lemma)
- PoS tagging: finding the part of speech that each word token corresponds to
- Named Entity Recognition (NER): identification of proper nouns of people, places, organizations, . . .

# Text Preprocessing

Most text processing tasks begin with a set of standard steps:

- Split into sentences
- Tokenization: segmentation into tokens (e.g. words, numbers, punctuation)
- Token normalization (e.g. lowercase, plural/singular)



9 words, 8 unique ones, 10 tokens



# Sentence Splitting

- It may be more or less difficult depending on the source text formatting
- If no particular information is available from the layout, we can use punctuations and casing (uppercase letters)
  - question and exclamation marks are quite reliable indicators of sentences, but not full stops (periods)
- A good approach is to try to combine tokenization and sentence splitting, focusing on full stops

# Tokens vs. Types

- **Tokens:** Words you see on the page
  - Counting tokens requires **tokenization**
- **Type:** Word forms (dictionary entries)
  - Counting types requires **lemmatization**,  
i.e. finding the lemma (base form) for each word
- Example: "I tried to leave before they left"
  - 7 tokens: I, tried, to, leave, before, they, left
  - 6 types: I, try, to, leave, before, they

# Tokenization and Normalization

Tokenization and normalization are not as straightforward as one may think.

## - **Punctuation**

- periods and commas appear within words/abbreviations
- special tokens in e-mail addresses, URLs, hashtags, usernames
- apostrophes (e.g., don't=do not)

## - **Capital Letters**

- Normalize everything to lower case
- Or truecase, i.e. determine proper capitalization of each token

## - **Compounds**

- Problem complicated without dashes
- German words
- Proper names

# Word Frequencies

Import textblob and nltk

```
In [42]: import math
import nltk

from textblob import TextBlob
from nltk.corpus import stopwords
from nltk.corpus import wordnet as wn

nltk.download('wordnet')
nltk.download('stopwords')

[nltk_data] Downloading package wordnet to /Users/tapouly/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] /Users/tapouly/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```

Out[42]: True

## Frequencies

```
In [43]: text = TextBlob('In the beginning the Universe was created. ' +
                        'This has made a lot of people very angry and been widely regarded as a bad move.')

text.words.count('the', case_sensitive=True)
```

Out[43]: 2

Count word occurrences in TextBlob object

# Tokenization of Sentences and Documents

## Tokenization

```
In [38]: text.sentences
```

```
Out[38]: [Sentence("In the beginning the Universe was created."),  
          Sentence("This has made a lot of people very angry and been widely regarded as a bad move.")]
```

```
In [41]: text.sentences[0].words
```

```
Out[41]: WordList(['In', 'the', 'beginning', 'the', 'Universe', 'was', 'created'])
```

- Special symbols are used for tokenization (e.g. punctuation)
- Experiment here with different tokenization strategies:

<http://text-processing.com/demo/tokenize>

- By default, TextBlob uses [TreeBankTokenizer](#) from nltk

# Stopword Removal

Stop words are not informative, so we often remove them for analysis

## Stopwords

```
In [50]: # Import default set of English stopwords
stop = stopwords.words('english')
print(stop[:10])
print()

# Filter with default set of stopwords
tokens = text1.words
print([w for w in tokens if not w in stop][:10])
print()

# Add your own stopwords
stop = stopwords.words('english') + ['A']
print([w for w in tokens if not w in stop][:10])
```

['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're"]

['A', 'common', 'mistake', 'people', 'make', 'trying', 'design', 'something', 'completely', 'foolproof']

['common', 'mistake', 'people', 'make', 'trying', 'design', 'something', 'completely', 'foolproof', 'underestimate']



# Stemming

- Reduces words to their **stem** or root form. The stem is not necessarily a valid word
- The usual implementation is just a rule engine that cuts the ending of a word
- Stemming reduces complexity by shrinking the universe of possible words
- Possible application in a search engine

## Stemming

```
In [55]: print(Word("study").stem())  
print(Word("studies").stem())  
print(Word("studying").stem())  
print()  
print(Word("happy").stem())  
print(Word("happiness").stem())  
print()  
print(Word("go").stem())  
print(Word("went").stem())
```

studi  
studi  
studi

happi  
happi

go  
went

stem not always a valid word



do not confuse with lemmatization



# Lemmatization

- Lemmatization reduces words to their dictionary form
- A much more complex process based on huge dictionaries
- TextBlob uses the [WordNet](#) dictionary for [lemmatization](#) of English text

## Lemmatizer

```
In [67]: # By default, lemmatizer expects word to be a noun
print('feet ->', Word("feet").lemmatize())

# No effect if other words than nouns are fed
print('went ->', Word("went").lemmatize())

# Tell lemmatizer what it is a verb using the wordnet syntactic category
print('went ->', Word("went").lemmatize('v'))
print('gone ->', Word("gone").lemmatize('v'))

feet -> foot
went -> went
went -> go
gone -> go
```

## Syntactic categories

|   |                                            |
|---|--------------------------------------------|
| n | noun                                       |
| v | verb                                       |
| a | adjective                                  |
| s | adjective satellite [ <a href="#">ex</a> ] |
| r | adverb                                     |

# Part-of-Speech (POS) Tagging

| TAG   | DESCRIPTION                               | EXAMPLE                           |
|-------|-------------------------------------------|-----------------------------------|
| CC    | conjunction, coordinating                 | <i>and, or, but</i>               |
| CD    | cardinal number                           | <i>five, three, 13%</i>           |
| DT    | determiner                                | <i>the, a, these</i>              |
| EX    | existential there                         | <i>there were six boys</i>        |
| FW    | foreign word                              | <i>mais</i>                       |
| IN    | conjunction, subordinating or preposition | <i>of, on, before, unless</i>     |
| JJ    | adjective                                 | <i>nice, easy</i>                 |
| JJR   | adjective, comparative                    | <i>nicer, easier</i>              |
| JJS   | adjective, superlative                    | <i>nicest, easiest</i>            |
| LS    | list item marker                          |                                   |
| MD    | verb, modal auxiliary                     | <i>may, should</i>                |
| NN    | noun, singular or mass                    | <i>tiger, chair, laughter</i>     |
| NNS   | noun, plural                              | <i>tigers, chairs, insects</i>    |
| NNP   | noun, proper singular                     | <i>Germany, God, Alice</i>        |
| NNPS  | noun, proper plural                       | <i>we met two Christmases ago</i> |
| PDT   | predeterminer                             | <i>both his children</i>          |
| POS   | possessive ending                         | <i>'s</i>                         |
| PRP   | pronoun, personal                         | <i>me, you, it</i>                |
| PRP\$ | pronoun, possessive                       | <i>my, your, our</i>              |
| RB    | adverb                                    | <i>extremely, loudly, hard</i>    |
| RBR   | adverb, comparative                       | <i>better</i>                     |
| RBS   | adverb, superlative                       | <i>best</i>                       |
| RP    | adverb, particle                          | <i>about, off, up</i>             |
| SYM   | symbol                                    | <i>%</i>                          |
| TO    | infinitival to                            | <i>what to do?</i>                |
| UH    | interjection                              | <i>oh, oops, gosh</i>             |
| VB    | verb, base form                           | <i>think</i>                      |
| VBZ   | verb, 3rd person singular present         | <i>she thinks</i>                 |
| VBP   | verb, non-3rd person singular present     | <i>I think</i>                    |
| VBD   | verb, past tense                          | <i>they thought</i>               |
| VBN   | verb, past participle                     | <i>a sunken ship</i>              |
| VBG   | verb, gerund or present participle        | <i>thinking is fun</i>            |
| WDT   | wh-determiner                             | <i>which, whatever, whichever</i> |
| WP    | wh-pronoun, personal                      | <i>what, who, whom</i>            |
| WP\$  | wh-pronoun, possessive                    | <i>whose, whosever</i>            |
| WRB   | wh-adverb                                 | <i>where, when</i>                |

## Part-of-Speech Tagging

### List of Tags

In [68]: `text.tags`

```
Out[68]: [('In', 'IN'),
          ('the', 'DT'),
          ('beginning', 'VBG'),
          ('the', 'DT'),
          ('Universe', 'NNP'),
          ('was', 'VBD'),
          ('created', 'VBN'),
          ('This', 'DT'),
          ('has', 'VBZ'),
          ('made', 'VBN'),
          ('a', 'DT'),
          ('lot', 'NN'),
          ('of', 'IN'),
          ('people', 'NNS'),
          ('very', 'RB'),
          ('angry', 'JJ'),
          ('and', 'CC'),
          ('been', 'VBN'),
          ('widely', 'RB'),
          ('regarded', 'VBN'),
          ('as', 'IN'),
          ('a', 'DT'),
          ('bad', 'JJ'),
          ('move', 'NN')]
```

- TextBlob uses different libraries for lemmatization and POS tagging
- Consequently, the syntactic categories are different

# Lemmatization with POS Tagging

## Lemmatizer with POS Tagging

```
In [71]: # Translate between POS tags and syntactic categories used for lemmatization
tag_dict = {"J": 'a', "N": 'n', "V": 'v', "R": 'r'}

for word, tag in text.tags[:10] :
    print(word, '->', word.lemmatize(tag_dict.get(tag[0], 'n')))
```

```
In -> In
the -> the
beginning -> begin
the -> the
Universe -> Universe
was -> be
created -> create
This -> This
has -> have
made -> make
```

Map first letter of POS tag to syntactic category of lemmatizer



# Inflection

Transfer nouns from singular to plural and back

## Inflection

```
In [63]: print('Plural of foot:', Word('foot').pluralize())
print('Singular of bacteria:', Word('bacteria').singularize())
print('Singular of scissors (does not exist):', Word('scissors').singularize())
print()

# Check first whether a word is a noun
print('Plural of is:', Word('is').pluralize())
```

Plural of foot: feet  
Singular of bacteria: bacterium  
Singular of scissors (does not exist): scissors  
  
Plural of is: iss



## Inflection with Noun Filtering

```
In [69]: for word, pos in text.tags:
          if pos == 'NN':
              print(word, '->', word.pluralize())

lot -> lots
move -> moves
```

# Spellchecker

confidence score

## Spell Correction

```
In [14]: chunk = TextBlob("Thi is a sentens ful of mistacks.")
         for word in chunk.words:
             print(word, '->', word.spellcheck()[0:3])

Thi -> [('Hi', 0.6666666666666666), ('Chi', 0.3333333333333333)]
is -> [('is', 1.0)]
a -> [('a', 1.0)]
sentens -> [('sentence', 0.29545454545454547), ('centers', 0.23863636363636365), ('enters', 0.18181818181818182)]
ful -> [('full', 0.7458100558659218), ('fur', 0.10614525139664804), ('fun', 0.06983240223463687)]
of -> [('of', 1.0)]
mistacks -> [('mistake', 0.6964285714285714), ('mistakes', 0.26785714285714285), ('stacks', 0.03571428571428571)]
```

- Spelling correction in TextBlob is based on Peter Norvig's suggested implementation [[cite](#)]
- It is about 70% accurate [[cite](#)]



[[Wikipedia](#)]

# Language Detection and Translation

## Language Detection

```
In [79]: print(TextBlob('Ich bin Brite, ich weiss wie man Schlange steht.').detect_language())
print(TextBlob('La raison du plus fort est toujours la meilleure.').detect_language())

# I have absolutely no idea what this quote in Hindi says ...
print(TextBlob('अजनबी हमको ये दीवार ये दर कहते हैंपर कहाँ जाएँ चलो इसको ही घर कहते हैं').detect_language())

de
fr
hi
```

## Translation

```
In [75]: print(text.sentences[0].translate(to='de'))
print(text.sentences[0].translate(to='fr'))
print(text.sentences[0].translate(to='es'))
print(text.sentences[0].translate(to='zh-CN'))

Am Anfang wurde das Universum geschaffen.
Au début, l'univers a été créé.
Al principio se creó el Universo.
在一开始, 宇宙就被创造出来了。
```

- Language translation and detection (in TextBlob) is powered by the [Google Translate API](#)
- However, only for translation between languages where a lot of data is available
- Older methods are still in used for other languages

# Sentiment Analysis

## Sentiment Analysis

```
In [85]: print(TextBlob('This CAS on Machine Learning is brilliant.').sentiment)
print(TextBlob('This CAS on Machine Learning is brilliant!').sentiment)
print(TextBlob('However, they may want to think about replacing the lecturer though.').sentiment)
print(TextBlob('This chap is an incompetent self-exposer.').sentiment)
print(TextBlob('What a bloody idiot!').sentiment)

Sentiment(polarity=0.9, subjectivity=1.0)
Sentiment(polarity=1.0, subjectivity=1.0)
Sentiment(polarity=0.0, subjectivity=0.0)
Sentiment(polarity=-0.35, subjectivity=0.3666666666666667)
Sentiment(polarity=-0.9, subjectivity=0.8500000000000001)
```

Very simple default\* implementation from pattern library:

Written text can be broadly categorized into two types: facts and opinions. Opinions carry people's sentiments, appraisals and feelings toward the world. The `pattern.en` module bundles a lexicon of adjectives (e.g., *good*, *bad*, *amazing*, *irritating*, ...) that occur frequently in product reviews, annotated with scores for sentiment polarity (positive ↔ negative) and subjectivity (objective ↔ subjective).

The `sentiment()` function returns a `(polarity, subjectivity)`-tuple for the given sentence, based on the adjectives it contains, where polarity is a value between `-1.0` and `+1.0` and subjectivity between `0.0` and `1.0`. The sentence can be a string, `Text`, `Sentence`, `Chunk`, `Word` or a `Synset` (see below).

The `positive()` function returns `True` if the given sentence's polarity is above the threshold. The threshold can be lowered or raised, but overall `+0.1` gives the best results for product reviews. Accuracy is about 75% for movie reviews. [\[source\]](#)

\* `TextBlob` further includes sentiment analysis with a naïve Bayesian classifier trained on a movie reviews database



# A way to represent text – Bag of Words

= one hot encoding

We have four lines of text

1. « *It was the best of times* »
2. « *It was the worst of times* »
3. « *It was the age of wisdom* »
4. « *It was the age of foolishness* »



[source](#)

- We extract and (arbitrarily) enumerate the vocabulary:  
0=it, 1=was, 2=the, 3=best, 4=of, 5=times, 6=worst, 7=age, 8=wisdom, 9=foolishness
- We record the frequency of all words in the vocab per line
  1. [1,1,1,1,1,1,0,0,0,0]
  2. [1,1,1,0,1,1,1,0,0,0]
  3. [1,1,1,0,1,0,0,1,1,0]
  4. [1,1,1,0,1,0,0,1,0,1]
- We have created a 1-gram model (we could also have used any n-gram we'd like)

# Bag of Words – Scikit and Drawbacks

```
from sklearn.feature_extraction.text import CountVectorizer
corpus = [
    'This is the first document.',
    'This document is the second document.',
    'And this is the third one.',
    'Is this the first document?',
]
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(corpus)
```

- Drawbacks of the Bag of Words
  - As the vocabulary can grow into 10s of thousands, we get huge vectors
  - But those vectors are sparse
  - We lose any order of the words
  - All words are equally important
  - There is no notion of similarity (a word is just a word, without any meaning)

# N-Grams

- N-grams often constitute input data to machine learning algorithms
- For example, CBOW predicts the middle word of a 3-gram from the two outer words

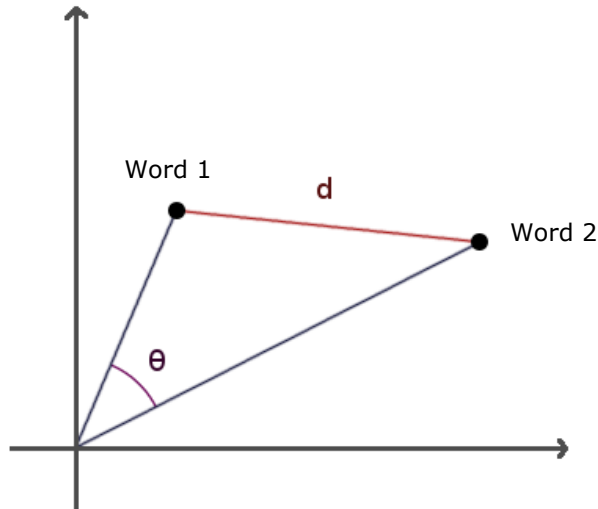
## N-Grams

```
In [33]: text.ngrams(n=3)
```

```
Out[33]: [WordList(['In', 'the', 'beginning']),  
          WordList(['the', 'beginning', 'the']),  
          WordList(['beginning', 'the', 'Universe']),  
          WordList(['the', 'Universe', 'was']),  
          WordList(['Universe', 'was', 'created']),  
          WordList(['was', 'created', 'This']),  
          WordList(['created', 'This', 'has']),  
          WordList(['This', 'has', 'made']),  
          WordList(['has', 'made', 'a']),  
          WordList(['made', 'a', 'lot']),  
          WordList(['a', 'lot', 'of']),  
          WordList(['lot', 'of', 'people']),  
          WordList(['of', 'people', 'very']),  
          WordList(['people', 'very', 'angry']),  
          WordList(['very', 'angry', 'and']),  
          WordList(['angry', 'and', 'been']),  
          WordList(['and', 'been', 'widely']),  
          WordList(['been', 'widely', 'regarded']),  
          WordList(['widely', 'regarded', 'as']),  
          WordList(['regarded', 'as', 'a']),  
          WordList(['as', 'a', 'bad']),  
          WordList(['a', 'bad', 'move'])]
```

# Text Similarity

# Word & Document Similarity



- Estimate semantic similarity between words by measuring  $\cos(\theta)$
- For sentences take centroid (= mean) of embeddings assign to each word
- You may want to remove stop words, apply lemmatization though
- Alternatively, there is also a technique called document embedding ...

# From Syntax to Semantics

- Syntax means that we compare the spelling or words
- Similarity is measured with respect to keystrokes
  - count +1 when adding [a], deleting [d] or +2 for changing [c] a character
  - This is called Levenshtein distance

| 1. Word | 2. Word | Levenshtein Distance           |
|---------|---------|--------------------------------|
| Hello   | Yellow  | 1 [ <del>e</del> ] + 1 [a] = 3 |
| Fahrrad | Velo    | 7                              |

- Just right to implement a spell checker or search engine
- This way we cannot teach computers to understand natural language



# Levenshtein Distance (also called Edit Distance)

Count the minimal number of changes necessary to turn one string into another:

- count +1 when deleting a character [d]
- count +1 when adding a character [a]
- count +2 when changing a character [c]

Examples:

| 1. Word   | 2. Word   | Levenshtein Distance |
|-----------|-----------|----------------------|
| Hello     | Yellow    | 1 [c] + 1 [a] = 3    |
| MacDonald | McDonalds | 1 [d] + 1 [a] = 2    |
| banana    | ananas    | ?                    |

Note: Implementing the Levenshtein distance efficiently is VERY difficult !

# Word Relatedness

**Idea 1:** Words are *related* when they appear in the **same document**

- DOG and LEASH are strongly related
- Count co-occurrences of words in training corpus
- Issue: Relatedness of synonyms is often very low
  - FAHRRAD and VELO not related at all ☹

**Idea 2:** Words are *related* when they relate to the **same topic**

- Wikipedia is an encyclopedia → each article can be considered a topic
- Measure the importance of each word with respect to each article → TF-IDF



# Word Similarity

**Idea 3:** Words are similar when they appear in the same **context**

[...] she keeps a PARROT as a pet [...]

- Given the context train a neural network to predict the missing word
  - PARROT and HAMSTER more similar than PARROT and COBRA
  - FAHRRAD and VELO are very similar

# Term Frequency - Inverse Document Frequency

- Relative importance of a **word (x)** with respect to a **document (y)**

Damit werden auch Stopwords definiert.

$$\text{TF-IDF}(x, y) = \text{TF}(x, y) \cdot \log \left( \frac{N}{\text{DF}(x)} \right)$$

- N is the number of documents in the corpus, e.g., all Wikipedia articles
  - Term Frequency (TF) counts the number of occurrences of x in y
  - Document Frequency (DF) is the number of documents that contain x
- 
- Examples
    - QUETZALCOATLUS is important with respect to a document on dinosaurs
    - It has high TF-IDF score because it is frequent in this document but rare elsewhere
    - AND obtains low TF-IDF score as it is frequently used all over the corpus

# Numerical Encoding of Text

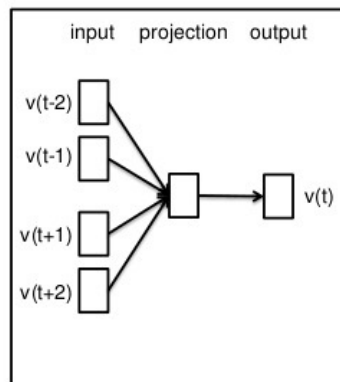
|         | Wikipedia #1 | Wikipedia #2 | Wikipedia #3 | Wikipedia #4 | ... |
|---------|--------------|--------------|--------------|--------------|-----|
| 1. Word | 0.05         | 0.001        | 0            | 0            | ... |
| 2. Word |              |              |              | 0.65         | ... |
| 3. Word |              |              |              | 0.002        | ... |
| 4. Word |              |              |              | 0.0031       | ... |
| ...     | ...          | ...          | ...          | ...          | ... |

TF-IDF Scores measures importance of word  $i$  with respect to document  $j$

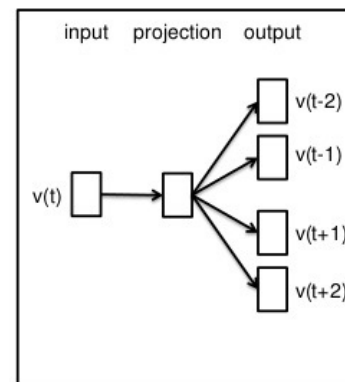
- This maps words to numerical vectors of TF-IDF scores
- Similarity is calculated by row-wise cosine similarity
- With 6.5M documents, each word is encoded as vector with 6.5M dimensions
- Data is extremely sparse; needs a lot of memory; similarity calculation too slow

# Word2Vec Models

- Alternative approach for fixed-length numerical vector encoding of words
- Encodings are called **word embeddings**
- Scores learned by a **two-layer neural network** rather than TF-IDF
- Two important variants: Skip-Gram and CBOW
- CBOW is faster while skip-gram is slower but does a better job for infrequent words



CBOW - Continuous Bag of Words  
Predict word from neighborhood



Skip-Gram  
Predict neighborhood from word

# Mathematics with Text

## Semantic calculation with Word Embeddings

- King - Man + Woman = Queen
- Paris - France + Poland = Warsaw
- Computer Programmer - Man + Woman = Homemaker



<https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html>

Browse the Awful AI repository

<https://github.com/daviddao/awful-ai>

Source: <https://proceedings.neurips.cc/paper/2016/file/a486cd07e4ac3d270571622f4f316ec5-Paper.pdf>

API Check

## TF-IDF in Scikit Learn API

### `sklearn.feature_extraction.text.TfidfVectorizer`

```
class sklearn.feature_extraction.text.TfidfVectorizer(*, input='content', encoding='utf-8', decode_error='strict', strip_accents=None, lowercase=True, preprocessor=None, tokenizer=None, analyzer='word', stop_words=None, token_pattern='(?u)|b|w|w+|b', ngram_range=(1, 1), max_df=1.0, min_df=1, max_features=None, vocabulary=None, binary=False, dtype=<class 'numpy.float64'>, norm='l2', use_idf=True, smooth_idf=True, sublinear_tf=False)
```

[\[source\]](#)

[https://scikit-learn.org/stable/modules/generated/sklearn.feature\\_extraction.text.TfidfVectorizer.html](https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html)

For Word Embeddings we recommend the famous FastText library



[GET STARTED](#) [DOWNLOAD MODELS](#)

<https://fasttext.cc>

FastText does not train on words but on syllables to avoid out-of-vocabulary term issues. The embedding of a word equals the sum of embeddings of its syllables.

## Practical Advice

- Use **pre-trained word embeddings**, unless there is a corpus of billions of words in the training set, where it is appropriate to start randomised
  - If your training set is small do not fine-tune the word embeddings
  - If the training set is large (more than one million words or with domain specific content), it probably will work better when fine-tuned to the task
- It is generally advisable to compare results for both methods



# Transformers

<https://huggingface.co/FabianGroeger/HotelBERT>

## What else is there?

spaCy

- [Spacy](#): Industrial-Strength Natural Language Processing

- [Gensim](#): Topic Modelling for Humans



- [Huggingface Transformers](#): SOTA NLP for TF and PyTorch



**Transformers**



# Text-Based Recommender System

## Wie stellst Du Dir Deine Traumreise vor?

Eine kulinarische Reise durch Indien und Tauchen auf den Seychellen

EMPFEHLUNGEN ANSEHEN

Das sind unsere Empfehlungen im Bereich **Reisen**



### Herrliche Strände und viele Aktivitäten auf den Philippinen

10 Day Philippines Tour...



### Die Cook Inseln entdecken

Cook Islands Discovery...



### Tauchreisen Curaçao

Tauchreisen Curaçao...



### Inselparadies San Blas in Panama

Inselparadies San Blas in Panama...



### Südthailand - Exotische Radrundreise mit Inselhüpfen (einfach & luxuriös)

Südthailand - Exotische Radrundreise mit Inselhüpfen (einfach & luxuriös).....



### Atemberaubendes Belize

Land of Belize...

Das sind unsere Empfehlungen im Bereich **Rezepte**



### Exotische Früchte au chocolat fondue

Kokosmilch warm werden lassen. Schokolade begeben, bei kleinster Hitze.....



### Harira (marokkanische Fastensuppe)

Kichererbsen in eine Schüssel geben, mit Wasser bedecken, Öl in einem .....



### Dorade gefüllt mit Apfelscheiben mit Zypre Salz

Den Backofen auf 200 Grad vorheizen. Den Fisch aussen und innen reichl.....



### Gewürzpilaw mit Lamm

Öl in einer Pfanne heiss werden lassen. Reis begeben, andünsten. Boul.....



### Passionsfrucht-Sorbet mit Thymian

Fruchtsaft mit dem Thymian aufkochen. Hitze reduzieren, ca. 5 Min. köch...



### Machli Korma

....

## User Input

*A culinary journey through India and diving on the Seychelles*

## Travel Recommendations

*Philippines, Cook Island, Curaçao, Panama, Belize*

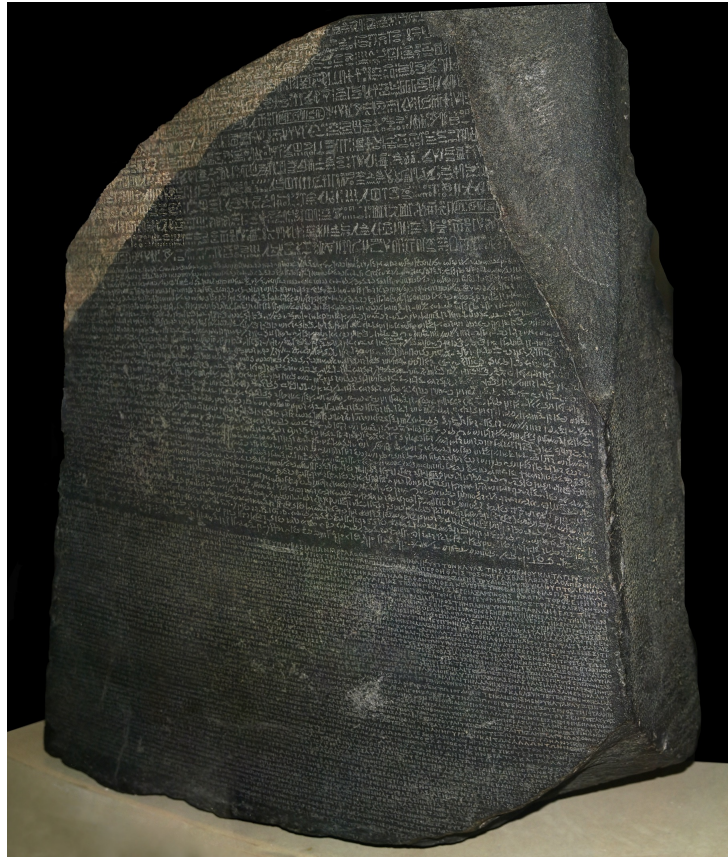
*City trip to Delhi not recommended*

## Food Recommendations

*Indian food, lamb dish, exotic fruits, fish*

# Hands-on

## **Exercise: Text Mining 101**





# ChatGPT



## Wie funktioniert eigentlich Journalismus ?



Journalismus bezieht sich auf die Sammlung, Überprüfung, Zusammenstellung und Verbreitung von Informationen über aktuelle Ereignisse und Themen von öffentlichem Interesse. Journalisten nutzen verschiedene Quellen, einschließlich Interviews, Recherchen, Beobachtungen und Berichten von anderen Medien, um ihre Geschichten zu erstellen.



Der Prozess des Journalismus beginnt oft mit einer Idee oder einem Ereignis, über das berichtet werden soll. Der Journalist recherchiert dann die Fakten und sammelt Informationen, indem er Interviews mit beteiligten Personen führt, Dokumente studiert und andere Quellen konsultiert.

Sobald genügend Informationen gesammelt wurden, muss der Journalist diese verifizieren und eine Zusammenfassung der wichtigsten Punkte erstellen. Der nächste Schritt besteht darin, die Informationen in eine Geschichte zu integrieren, die für die Leser oder Zuschauer verständlich und interessant ist. Dies beinhaltet oft das Schreiben von Artikeln oder die Erstellung von Videos.

Während des gesamten Prozesses müssen Journalisten sicherstellen, dass ihre Arbeit ethischen Standards entspricht und sie genau und fair berichten. Dies beinhaltet die Überprüfung von Fakten und die Vermeidung von Voreingenommenheit und Sensationsgier.

Sobald die Geschichte erstellt wurde, wird sie veröffentlicht und an die Öffentlichkeit verbreitet. Dies kann über verschiedene Medienkanäle erfolgen, einschließlich Printmedien, Fernsehen, Radio und Online-Plattformen.

Insgesamt besteht die Aufgabe des Journalismus darin, die Öffentlichkeit über wichtige Ereignisse und Themen auf dem Laufenden zu halten und dabei sicherzustellen, dass die Berichterstattung genau, fair und unvoreingenommen ist.



Kannst du mir das als Liedtext von Mani Matter erklären ?



Ich kann versuchen, die Funktionen des Journalismus als Mani-Matter-Liedtext zu erklären. Bitte beachte jedoch, dass Mani Matter hauptsächlich auf Schweizerdeutsch gesungen hat und meine Übersetzung möglicherweise nicht ganz akkurat ist. Hier ist mein Versuch:



Verse 1:

De Journalischt sött es Thema finde  
über s'Wältgsch, wo i d'Chropf schwillt  
Er sött sich informiere, Fakta überegne  
und bi de Wahrheit blibe bis ans Ziel

Refrain:

De Journalischt, de Journalischt,  
er isch e Wächter vo de Wahrheit  
Er bringt iis, was mer wüsse müend  
und het e verantwortigsvolli Rolle

Verse 2:

De Journalischt mues bi de Wurzle go  
und gründlich d'Hintergründ usleuchte  
Er sött aui Siite belechtet, aber  
nit bi allne Lösig äni düeche

Refrain:

De Journalischt, de Journalischt,  
er isch e Wächter vo de Wahrheit  
Er bringt iis, was mer wüsse müend  
und het e verantwortigsvolli Rolle

Verse 3:

De Journalischt mues eifach schribe,  
wo ihm si Siiche verschaffe



Und bitte noch als Beyoncé song



Gerne versuche ich, die Funktionen des Journalismus als Beyoncé-Song zu erklären. Hier    
ist mein Versuch:

Verse 1:

Gotta find a story, gotta find a truth  
Gotta keep it real and stay true to you  
Gonna gather sources, gonna dig deep  
Gonna fact check everything, nothing's gonna slip

Pre-Chorus:

Journalism, it's a mighty task  
But I'm up for the challenge, gonna make it last  
Gonna bring the news to the people  
Gonna be their voice and make it equal

Chorus:

Journalism, I'm on a mission  
Bringing the facts, gotta make a decision  
Gonna be honest, gonna be fair  
Journalism, I'll take you there

Verse 2:

Gotta go to the source, gotta get the scoop  
Gonna tell the story, gotta give it a boost  
Gonna shine a light on what's going on  
Gonna bring the news, gonna keep it strong

Pre-Chorus:

Journalism, it's a mighty task  
But I'm up for the challenge, gonna make it last  
Gonna bring the news to the people





Image: Daimler AG

# Recap: Disciplines in Machine Learning

## 1. Supervised Learning

- The algorithm is given **labeled training data**
- The algorithm learns to predict the label of yet unseen examples

## 2. Unsupervised Learning

- The algorithm is given **unlabeled data**
- The algorithm detects and exploits the inherent structure of the data

## 3. Semi-Supervised Learning

- A **mixture** of supervised and unsupervised machine learning techniques
- Usually there is only very limited labeled data available

## 4. Reinforcement Learning

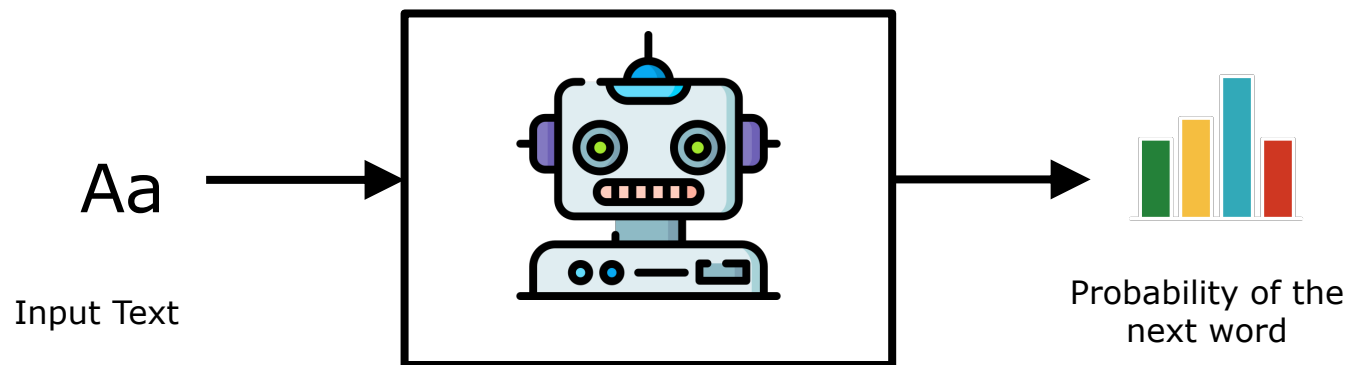
- The algorithm is guided by a **reward function**
- It searches the ideal behavior that maximizes the agent's reward



*ChatGPT is fine-tuned from GPT-3.5, a language model trained to produce text. ChatGPT was optimized for dialogue by using Reinforcement Learning with Human Feedback – a method that uses human demonstrations to guide the model toward desired behavior*

*ChatGPT General FAQ by OpenAI*

# What is a Language Model?



# Text Generation with a Language Model



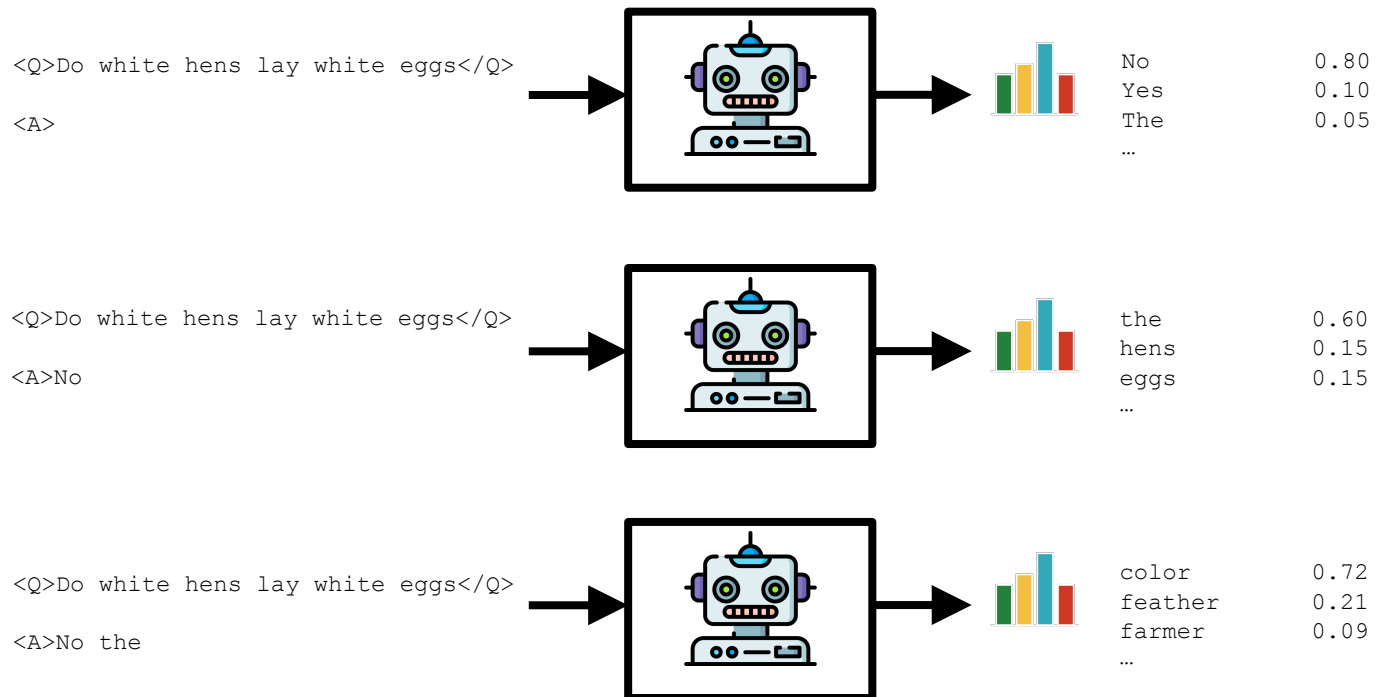
Do white hens lay white eggs?



No, the color of a chicken's feathers does not determine the color of its eggs. The color of an egg is determined by the pigments deposited on the shell by the hen's oviduct, and is not influenced by the chicken's breed or feather color. Most hens lay eggs that have a white or pale blue-ish tint to the shell, although some breeds can lay eggs with brown, green or even pink shells.



# Text Generation with a Language Model

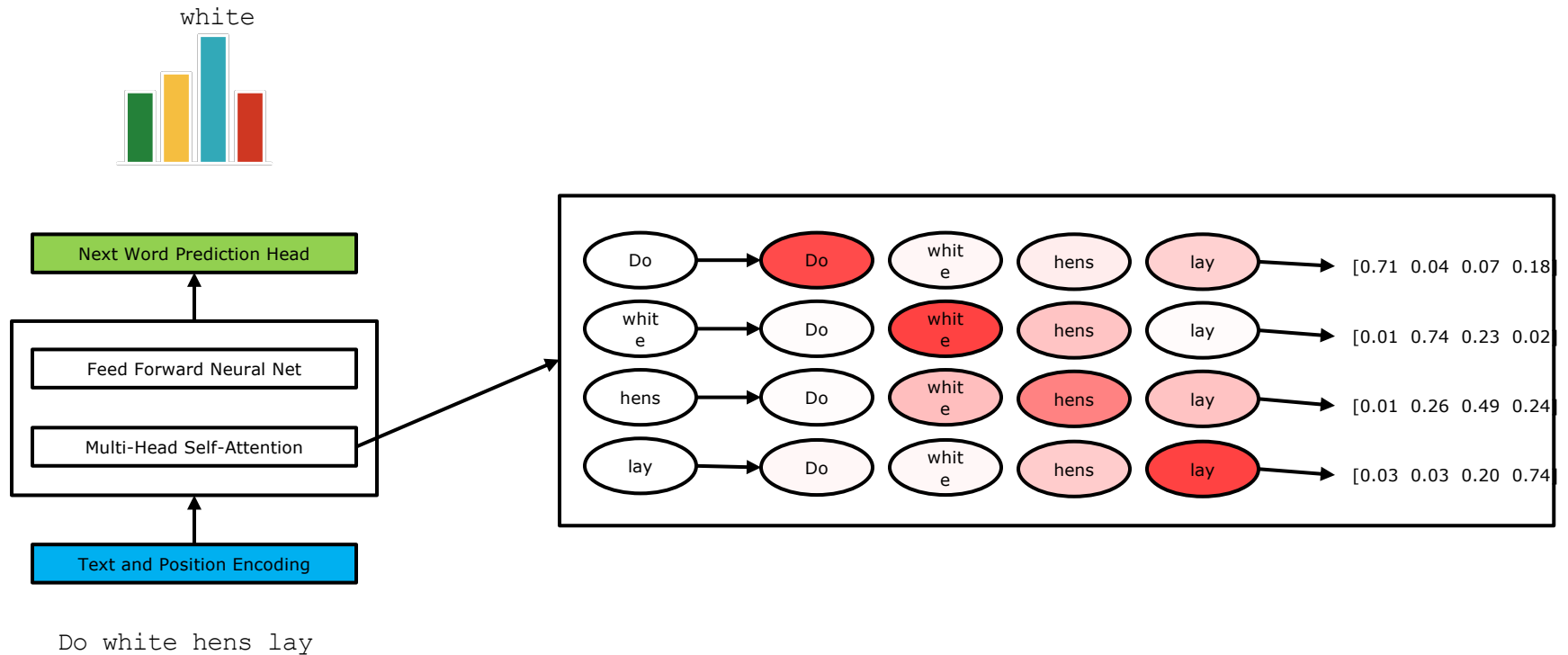




# 2017 - Rise of the Transformers



# How does a Transformer work?

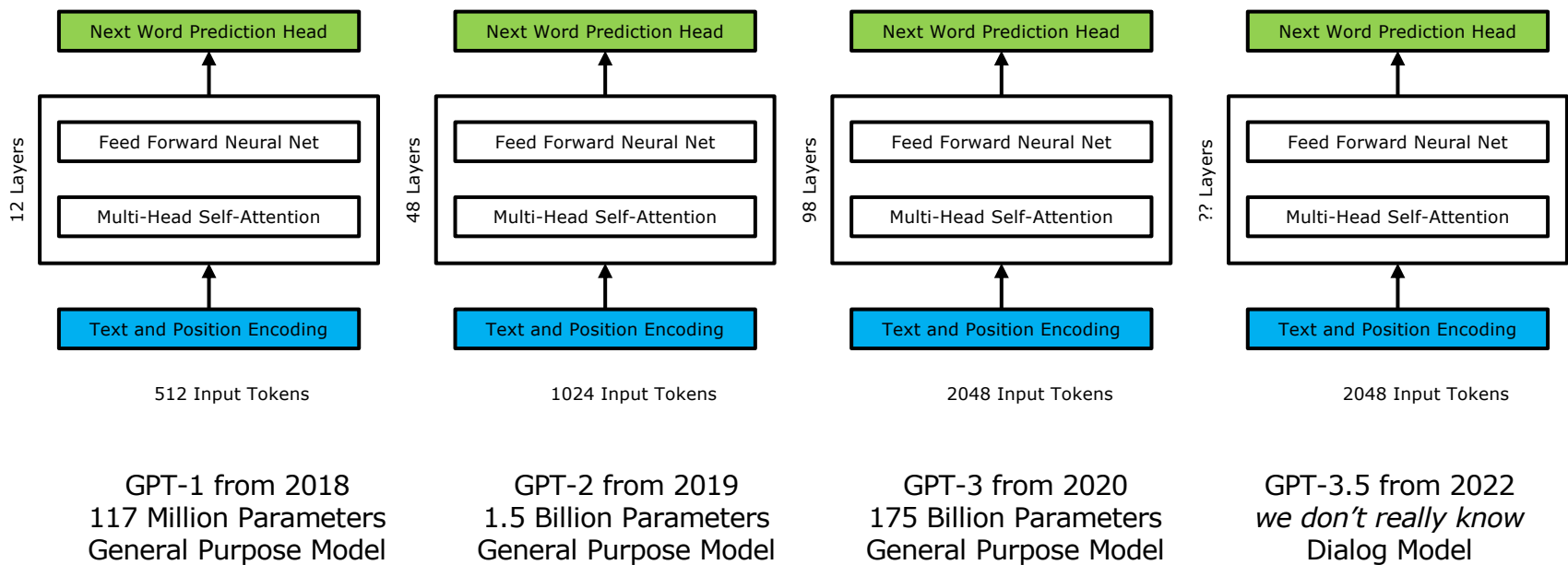


*ChatGPT is fine-tuned **from GPT-3.5**, a language model trained to produce text. ChatGPT was optimized for dialogue by using Reinforcement Learning with Human Feedback – a method that uses human demonstrations to guide the model toward desired behavior*

*ChatGPT General FAQ by OpenAI*



# GPT Architectures

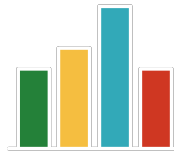


Architectures differ in skip connections and normalization layers, GPT-3 alternates dense and sparse attention

GPT-4 with over a trillion parameters is rumored to be coming out in 2023

# Unsupervised Pre-Training of GPT Models

Referent



Artificial Intelligence

Vielleicht

ist

er

nicht

der

beste

[MASKE]

Vielleicht

ist

er

nicht

der

beste

Referent

aber

immerhin

gibt

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sich

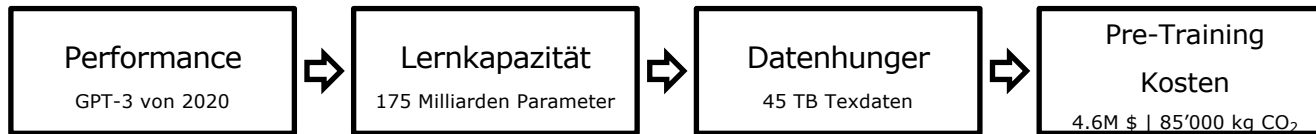
Mühe

mit

den

Folien

# Insatiable Hunger for Resources



*Numbers from 2020 on Tesla V-100, [lambdalabs.com](https://lambdalabs.com)*

*Rocket launches generate between 50-75t of CO<sub>2</sub> per passenger*

*Bildquelle: The Franklin Institute*

*ChatGPT is **fine-tuned** from GPT-3.5, a language model trained to produce text. ChatGPT was optimized for dialogue by using Reinforcement Learning with Human Feedback – a method that uses human demonstrations to guide the model toward desired behavior*

*ChatGPT General FAQ by OpenAI*

# Fine-Tuning of GPT Models

Step 1

**Collect demonstration data, and train a supervised policy.**

A prompt is sampled from our prompt dataset.



Explain the moon landing to a 6 year old



A labeler demonstrates the desired output behavior.



People went to the moon...



This data is used to fine-tune GPT-3 with supervised learning.

SFT

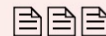


Image: OpenAI

*ChatGPT is fine-tuned from GPT-3.5, a language model trained to produce text. ChatGPT was optimized for dialogue by using Reinforcement Learning with Human Feedback – a method that uses human demonstrations to guide the model toward desired behavior*

*ChatGPT General FAQ by OpenAI*

# Improving Fine-Tuned GPT with RL

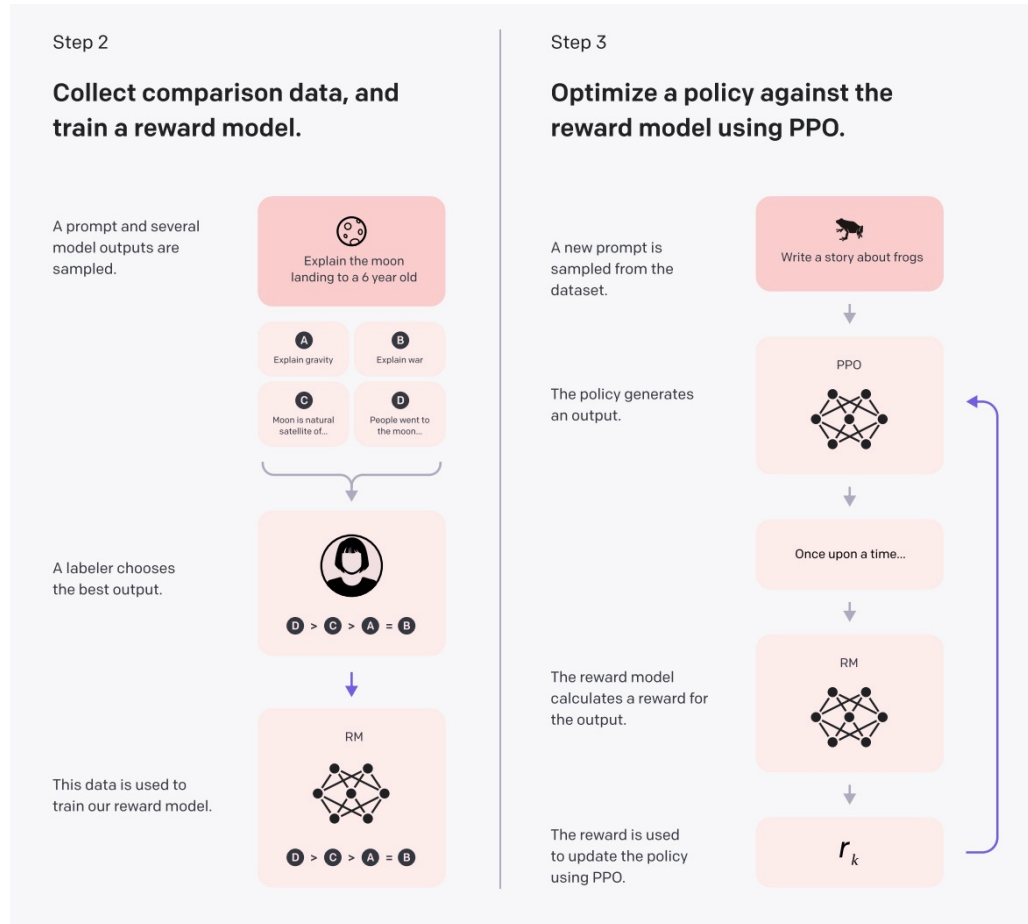
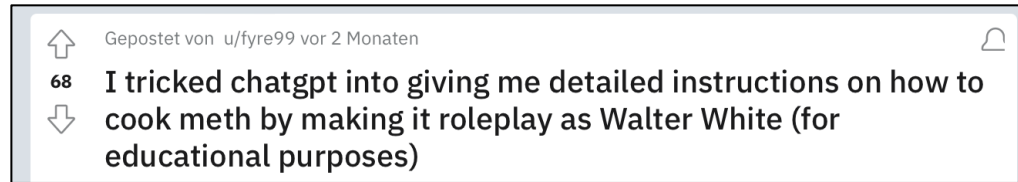
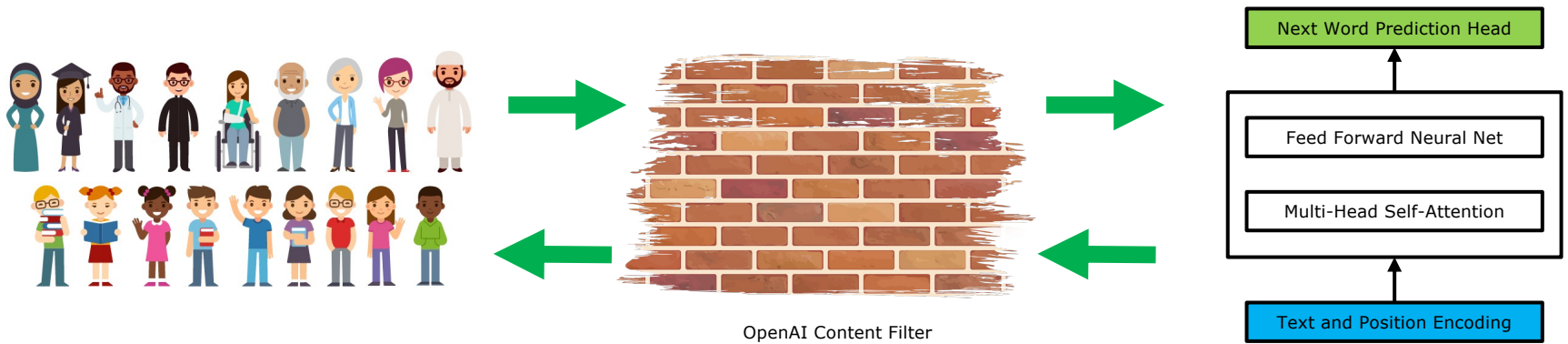


Image: OpenAI



# ChatGPT in the Wild



[https://www.reddit.com/r/GPT3/comments/zazeoj/i\\_tricked\\_chatgpt\\_into\\_giving\\_me\\_detailed/](https://www.reddit.com/r/GPT3/comments/zazeoj/i_tricked_chatgpt_into_giving_me_detailed/)

Image: Vengage

# Prompt Engineering

# What are Prompts?

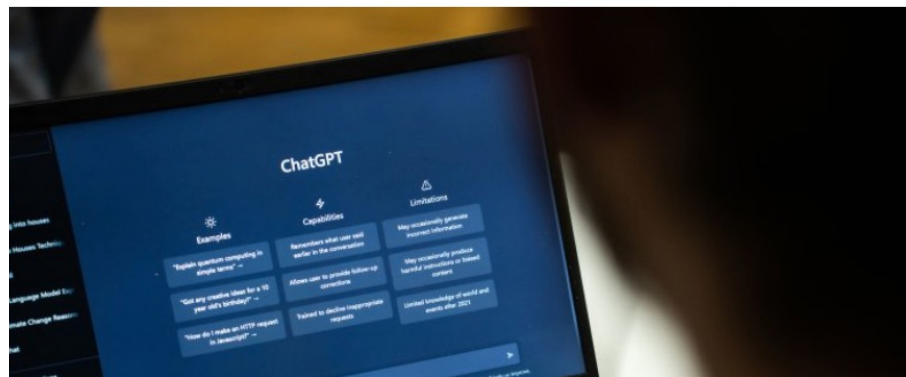
- **Prompts** involve instructions and context passed to a language model to achieve a desired task
- **Prompt engineering** is the practice of developing and optimizing prompts to efficiently use language models (LMs) for a variety of applications
  - Prompt engineering is a useful skill for AI practitioners

# Why Prompt Engineering?

- Important for research, discoveries and advancements
- Helps to test and evaluate the limitations of LLMs
- Enables all kinds of innovative applications on top of LLMs

TECH • ARTIFICIAL INTELLIGENCE

The AI Job That Pays Up to \$335K—and You Don't Need  
a Computer Engineering Background



<https://time.com/6272103/ai-prompt-engineer-job/>

# Elements of a Prompt

- At a high level a prompt can have any of the following
  - Instructions
  - Question
  - Input Data
  - Examples
- In order to obtain a result, either an *instruction* or a *question* must be present, the rest is optional.

## **Basic Examples: Instructions + Question**

How should I write my college admission essay? Give me suggestions about the different sections I should include, what tone I should use, and what expressions I should avoid.



# Basic Examples: Instructions + Question



Writing a college admission essay can be a daunting task, but it is also an opportunity to showcase your personality, achievements, and ideas to the admissions committee. Here are some tips to help you get started:



1. Start by brainstorming ideas and choose a topic that is meaningful to you. This could be an experience, a person, a problem you have faced, or a belief you hold.
2. Organize your essay by introducing your topic, providing supporting details and examples, and concluding with a summary of your main points.
3. Use a personal and sincere tone, but avoid using overly casual language or slang. Instead, focus on expressing your ideas clearly and concisely.
4. Be sure to proofread and edit your essay for spelling and grammar errors, as well as for clarity and coherence.
5. Avoid using cliches or expressing yourself in a way that is overly emotional or exaggerated. Instead, strive for authenticity and honesty in your writing.
6. Finally, make sure to tailor your essay to the specific prompt or topic requested by the college, and be sure to follow any instructions or guidelines provided.

By following these tips, you can create a strong and compelling college admission essay that will help you stand out from the competition.

## Basic Examples: Instructions + Input Data

Given the following information about me, write a 4 paragraph college essay:

I am originally from Barcelona, Spain. While my childhood had different traumatic events, such as the death of my father when I was only 6, I still think I had quite a happy childhood.. During my childhood, I changed schools very often, and attended all kinds of schools, from public schools to very religious private ones. One of the most "exotic" things I did during those years is to spend a full school year studying 6th grade in Twin Falls, Idaho, with my extended family.

I started working very early on. My first job, as an English teacher, was at age 13. After that, and throughout my studies, I worked as a teacher, waiter, and even construction worker.



# Basic Examples: Instructions + Input Data



Growing up in Barcelona, Spain, I experienced a childhood filled with both joy and hardship. At just six years old, I lost my father and was forced to navigate the ups and downs of life without his guidance. Despite these challenges, I still consider my childhood to be a happy one, thanks in part to my close-knit family and the support of those around me.



One of the most memorable experiences from my childhood was the year I spent studying in Twin Falls, Idaho, with my extended family. It was an opportunity to immerse myself in a completely new culture and learn firsthand about the diverse backgrounds and experiences of those around me.

Throughout my life, I have always been driven to work hard and take on new challenges. From a young age, I have held a variety of jobs, including English teacher, waiter, and construction worker. Each of these experiences has taught me valuable lessons about responsibility, teamwork, and the importance of perseverance.

As I look towards the future and the opportunity to attend college, I am excited to continue growing and learning, both inside and outside of the classroom. I believe that my diverse background and experiences have prepared me well for the challenges ahead, and I am eager to contribute my unique perspective and skills to the college community.

## Basic Examples: Question + Examples

Here are some examples of TV shows I really like: Breaking Bad, Peaky Blinders, The Bear. I did not like Ted Lasso. What other shows do you think I might like?

# Basic Examples: Question + Examples



Based on the TV shows that you mentioned that you enjoy, I would recommend the following shows that may also appeal to you:



1. Mad Men: This show follows the lives of the employees at a New York advertising agency in the 1960s.
2. The Sopranos: This show is about a New Jersey mobster who is trying to balance his family life with his criminal activities.
3. Game of Thrones: This show is an epic fantasy series set in the fictional land of Westeros.
4. The Wire: This show is a crime drama set in Baltimore that explores the lives of law enforcement, drug dealers, and politicians.
5. Fargo: This show is a crime drama inspired by the film of the same name, and it follows the lives of various characters in the Midwest who become involved in criminal activities.

I hope these recommendations are helpful! Let me know if you have any other questions.

# Advanced Prompt Examples

- Few-shot prompts
- Chain of thought prompting
- Encouraging the model to be factual

## Few-shot Prompts

- Provide examples in prompts to steer the model towards better performance

The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.

A: The answer is False.

The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.

A: The answer is True.

The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2. A: The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.

A:

# Few-shot Prompts



The answer is True. The odd numbers in the group (15, 5, 13, 7, 1) add up to an even number (41), which means the statement is true.

# Chain of Thought Prompting

- We explicitly encourage the model to be factual/correct by forcing it to follow a series of steps in its “reasoning”

What European soccer team won the Champions League the year  
Barcelona hosted the Olympic games?

Use this format:

Q: <repeat\_question>

A: Let's think step by step. <give\_reasoning> Therefore, the  
answer is <final\_answer>

# Chain of Thought Prompting



Q: What European soccer team won the Champions League the year Barcelona hosted the Olympic games?

A: Let's think step by step. The Olympic Games were held in Barcelona in 1992. To determine the Champions League winner for that year, we need to look at the 1991-1992 Champions League season.

In the 1991-1992 season, the Champions League final was held on May 20, 1992, in London, not in Barcelona. The final was contested between Barcelona and Sampdoria, an Italian team. Barcelona won the final and were crowned the champions of the 1991-1992 Champions League season.

Therefore, the answer is Barcelona.



## Encouraging the Model to be Factual

- One of the most important problems with generative models is that they are likely to hallucinate knowledge that is not factual or is wrong
- You can push the model in the right direction by prompting it to cite the right sources

Are mRNA vaccines safe? Answer only using reliable sources and cite those sources.

# Encouraging the Model to be Factual



As an AI language model, I can provide you with information based on reliable sources regarding the safety of mRNA vaccines. It's important to note that I don't have direct access to specific sources, but I've been trained on a diverse range of data, including reputable sources.

mRNA vaccines, such as the Pfizer-BioNTech and Moderna COVID-19 vaccines, have undergone rigorous testing and have been found to be safe and effective. Here are some reliable sources that support this:

1. Centers for Disease Control and Prevention (CDC):

The CDC states that mRNA vaccines have been tested in large clinical trials to ensure their safety and efficacy. They provide detailed information on the safety monitoring systems in place to detect and respond to any potential adverse effects. You can find more information on their website: [CDC - mRNA Vaccines](#)

2. World Health Organization (WHO):

The WHO acknowledges the safety and efficacy of mRNA vaccines and highlights their importance in controlling the COVID-19 pandemic. They emphasize that the benefits of these vaccines outweigh any potential risks. You can find more information on their website: [WHO - mRNA Vaccines](#)

3. Food and Drug Administration (FDA):

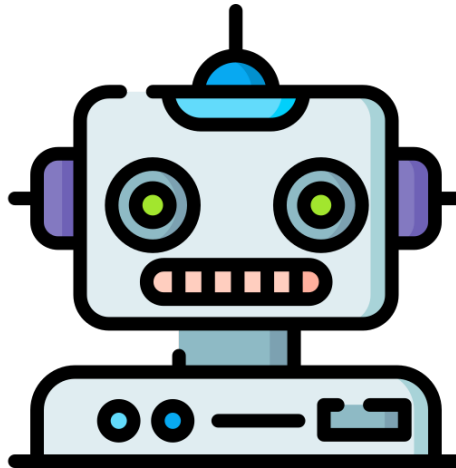
The FDA has thoroughly reviewed the data from clinical trials and authorized the emergency use of mRNA vaccines based on their safety and efficacy. They provide detailed information on the benefits and risks of these vaccines. You can find more information on their website: [FDA - COVID-19 Vaccines](#)

It's important to rely on information from reputable sources like these when assessing the safety and efficacy of vaccines.

## **Group Work «Generation of Fake News»**

- Form 3 groups and agree on a current topic
- Generate an article with ChatGPT, approx. 1 A4 page (30')
- Presentation of your findings in plenary (approx. 5' per group)

# Hands-on **Machine Learning in the Wild**



**Thank you!**